

Fenil Miteshkumar Modi

Ahmedabad, Gujarat, India | +91 7874160204 | fenilmmodi@gmail.com
[linkedin.com/in/fenilmmodi823](https://www.linkedin.com/in/fenilmmodi823) | github.com/fenilmmodi823

PROFILE

Computer Engineering graduate (degree conferral pending) seeking research positions in **space situational awareness** and **large-scale scientific data processing**. First author and presenter of an IEEE SMC-sponsored paper on probabilistic conjunction assessment, with experience across the full analysis chain — acquisition, preprocessing, exploratory analysis, probabilistic modelling, uncertainty quantification and validation. Currently Lead Backend & Data Engineer while leading a six-member orbital-analysis research team. Available for international relocation.

PUBLICATION

Probabilistic Space Debris Conjunction Assessment using Machine Learning and Covariance Intersection Analysis	Accepted
First author and presenter, with S. V. Khara, G. D. Tivari, J. Patel, P. Patel and G. Kumawat. ICSSIT 2026, Paper ID 1849 — technically co-sponsored by the IEEE Systems, Man and Cybernetics Society. Presented online, 28–30 July 2026; awaiting IEEE Xplore indexing. - Showed that a nominal miss distance can remain statistically consistent with collision, as measured by overlap of 3σ covariance ellipses projected into the B-plane, by fusing state estimates of unknown correlation through covariance intersection and screening candidate conjunctions with machine learning.	

EXPERIENCE

Hubbl — Technology Community Platform	India
<i>Lead Backend & Data Engineer Full-time</i>	<i>Jul 2026 – Present</i>
- Defined the backend and data architecture for a platform consolidating four regional technology user groups into one ecosystem, as measured by 8+ core backend services and the PostgreSQL schema now underpinning all community, event and speaker records, by setting service boundaries and data contracts at platform inception. - Turned fragmented per-group records into a single queryable dataset, as measured by four previously separate user-group datasets normalised into one relational schema, by designing the data model and building the ingestion path. - Kept community and event data current without manual entry, as measured by mail-synchronisation and event-scraping jobs running unattended on schedule, by building Node-based background workers against external community and event sources. - Delivered the registration data pipeline for a 120-capacity national analytics event with the India AI + Tableau User Group and Queen’s University Belfast (India GIFT City), as measured by a fully pre-registered cohort requiring no on-the-spot fallback, by owning participant records and registration analytics end to end.	
ORCAS — Orbital Risk and Conjunction Assessment System	Ahmedabad, India
<i>Team Lead & Lead Analyst Silver Oak University</i>	<i>Jan 2024 – Present</i>
- Cut catalogue preparation from a manual multi-hour task to a single automated pass, as measured by end-to-end runtime over a $\sim 10,000$-object TLE catalogue, by building a Python ingestion pipeline that validates checksums, resolves epoch drift and de-duplicates records. - Narrowed the conjunction search space before expensive computation, as measured by the count of candidate object pairs surviving each filter stage, by implementing a coarse-to-fine screening cascade using perigee–apogee and geometric pre-filters. - Replaced single point estimates with quantified confidence, as measured by 95% confidence intervals on probability of collision, by running Monte Carlo sampling over positional covariance in a batch-parallel workload. - Established propagator correctness before publishing any result, as measured by position residuals against published reference ephemerides across a multi-day horizon, by cross-validating the SGP4 implementation against independent reference outputs. - Kept a six-member team shipping reproducibly, as measured by full reconstruction of every published figure from versioned source, by instituting Git-based code review and parameterised analysis notebooks.	

SELECTED PROJECTS

Quantum-Random Key Generation — Statistical Evaluation <i>Python, Qiskit, SciPy</i>	2023
- Determined whether quantum-derived keys outperform pseudo-random streams, as measured by pass rates across the 15 statistical tests of NIST SP 800-22, by designing a controlled comparison over matched-length key sets. - Characterised diffusion behaviour with stated significance rather than headline claims, as measured by χ^2 goodness-of-fit and avalanche-effect distributions, by implementing entropy and bit-correlation analysis in NumPy/SciPy.	

EDUCATION

Silver Oak University	Ahmedabad, India
<i>B.Tech., Computer Engineering — coursework completed, degree conferral pending</i>	<i>Aug 2022 – Aug 2026</i>
Coursework: Statistical Methods, Machine Learning, Data Structures & Algorithms, Cryptography, Computer Networks, Database Systems, Scientific Computing.	

TECHNICAL SKILLS

Scientific Computing & Analysis: Python (NumPy, SciPy, Pandas), exploratory data analysis, time-series analysis, hypothesis testing, Monte Carlo methods, uncertainty quantification, numerical integration

Large-Scale Data Processing: batch pipelines, catalogue-scale datasets, scheduled background jobs, performance profiling, parallel workloads, Linux/CLI, high-performance computing concepts

Orbital & Space Data: SGP4/SDP4, TLE handling, Skyfield, ephemeris validation, conjunction screening, covariance intersection, B-plane geometry, coordinate frame transformations

Machine Learning: Scikit-learn, TensorFlow, PyTorch; cross-validation, residual analysis, model selection

Backend & Data Engineering: REST API design, schema design, query optimisation, JWT authentication, Node.js, Express.js, PostgreSQL, Prisma, FastAPI, MongoDB, React

Research Practice: Git/GitHub, reproducible analysis, LaTeX, Jupyter, Matplotlib, Tableau, technical documentation

ADDITIONAL

Languages: English (full professional), Hindi (native), Gujarati (native)

Interests: space situational awareness, orbital debris environment modelling, Kessler syndrome dynamics